

The Regularity Lemma for Stable Graphs and its Applications in Property Testing

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October 5, 2025

Table of contents

- ➊ Introduction to Szemerédi's Regularity Lemma
- ➋ Stable Graphs and the Stable Regularity Lemma
- ➌ Sketch of the Proof of the Stable Regularity Lemma
- ➍ Property Testing of H -freeness in Stable Graphs
- ➎ Conclusion and Future Work

Introduction to Szemerédi's Regularity Lemma

The idea of the SzRL is that all graphs can be partitioned in such a way that most pairs of parts behave in a quasi-random way, a property which is formalized as *regularity*.

Definition

Given $\epsilon > 0$ and a graph G , a pair of (not necessarily disjoint) subsets of vertices $A, B \subseteq G$ is said to be ϵ -regular if for all $A' \subseteq A$ and $B' \subseteq B$ such that $|A'| \geq \epsilon|A|$ and $|B'| \geq \epsilon|B|$, we have

$$|d(A', B') - d(A, B)| \leq \epsilon.$$

Introduction to Szemerédi's Regularity Lemma

When most of the pairs satisfy this property, we say that the partition is *regular*.

Definition

Let G be a graph and let $\epsilon > 0$. An ϵ -regular partition of G is a partition of its vertex set into k parts $\{A_1, \dots, A_k\}$ with remainder set B such that:

- $|B| \leq \epsilon|G|$, and may be empty.
- All but at most ϵk^2 of the pairs (A_i, A_j) with $1 \leq i < j \leq k$ are ϵ -regular.

Introduction to Szemerédi's Regularity Lemma

Theorem (Szemerédi's Regularity Lemma, [?])

For every $\epsilon > 0$ and every positive integer m , there exists a positive integer $M = M(\epsilon, m)$ such that every graph with at least m vertices admits an ϵ -regular partition $\{A_1, \dots, A_k\}$ and remainder B with $m \leq k \leq M$, with $|A_1| = \dots = |A_k|$.

Crucially, the theorem proves that the number of parts k does not grow with the size of the graph, but only depends on ϵ (and m).

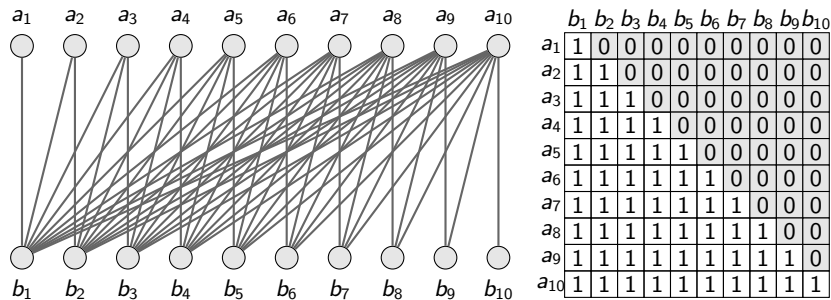
Introduction to Szemerédi's Regularity Lemma

However, the lemma has two major drawbacks:

- 1 The bound on the number of parts M is a *tower-type* function of ϵ , i.e. a power-tower of height polynomial in $1/\epsilon$. This is unavoidable, as shown by Gowers [?].
- 2 The partition may contain some irregular pairs. This is also unavoidable, and the half-graph H_n is an example.

Stable Graphs and the Stable Regularity Lemma

Stable graphs are those that do not contain a *half-graph* as a bi-induced subgraph.



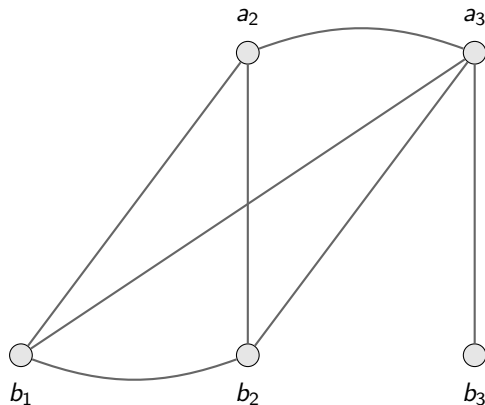
Stable Graphs and the Stable Regularity Lemma

Definition

Let G be a graph. We say that G has the *k -order property* if there exist two sequences of vertices $\langle a_i \mid i \in \{1, \dots, k\} \rangle$ and $\langle b_i \mid i \in \{1, \dots, k\} \rangle$ such that for all $i, j \leq k$, $a_i R b_j$ if and only if $i \geq j$. Otherwise, we say that G has the *non- k -order property* or that G is *k -stable*.

The two sequences need not to be disjoint, and any combination of edges and non-edges may appear within the vertices of each sequence.

Stable Graphs and the Stable Regularity Lemma



	a_2	a_3	b_1	b_2	b_3
a_2	0	1	1	1	0
a_3	1	0	1	1	1
b_1	1	1	0	1	0
b_2	1	1	1	0	0
b_3	0	1	0	0	0

Stable Graphs and the Stable Regularity Lemma

Theorem (Stable Regularity Lemma, [?])

Each k -stable graph G admits an equitable partition of its vertex set with at most $c(k) \cdot (1/\epsilon)^{2^{k+1}}$ parts such that ALL pairs of parts are ϵ -regular and have density either less than ϵ or greater than $1 - \epsilon$.

Notably both limitations of the SzRL are overcome: the bound on the number of parts is polynomial in $1/\epsilon$, and there are no irregular pairs.

Sketch of the Proof of the Stable Regularity Lemma

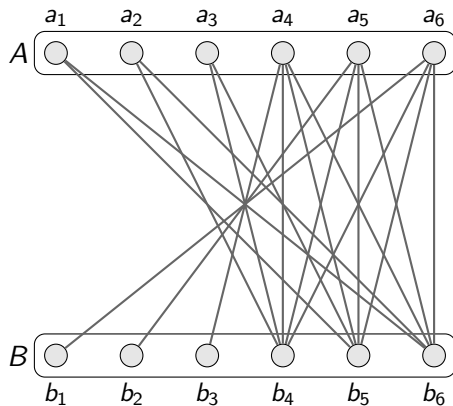
Definition (Good, [?])

Let G be a finite graph with the non- k_* -property. We say that $A \subseteq G$ is ϵ -good when for every $b \in G$ the truth value $t = t(b, A) \in \{0, 1\}$ satisfies $|\overline{B}_{A,b}| = |\{a \in A \mid aRb \neq t\}| < \epsilon|A|$.

Definition (Excellent, [?])

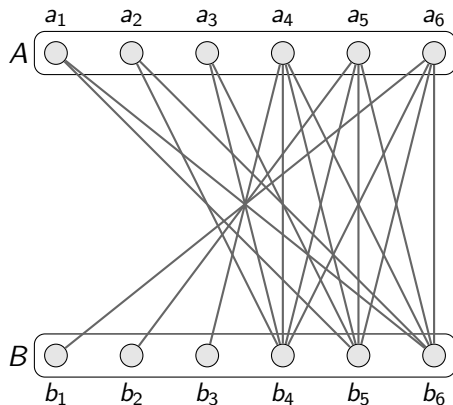
Let G be a finite graph with the non- k_* -property. We say that $A \subseteq G$ is (ϵ, ζ) -excellent when A is ϵ -good and, if B is ζ -good, then the truth value $t = t(B, A)$ satisfies $|\{a \in A \mid t(a, B) \neq t(A, B)\}| < \epsilon|A|$. In particular, we say A is ϵ -excellent if A is (ϵ, ϵ) -excellent.

Sketch of the Proof of the Stable Regularity Lemma



	b_1	b_2	b_3	b_4	b_5	b_6
a_1	0	0	0	0	1	1
a_2	0	0	0	1	0	1
a_3	0	0	0	1	1	0
a_4	0	0	1	1	1	1
a_5	0	1	0	1	1	1
a_6	1	0	0	1	1	1

Sketch of the Proof of the Stable Regularity Lemma



	b_1	b_2	b_3	b_4	b_5	b_6
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a_5	0	1	0	1	1	1
a_6	1	0	0	1	1	1

Do excellent subsets always exist?

Sketch of the Proof of the Stable Regularity Lemma

Definition (k -tree)

A k -tree in G is an ordered pair $H = (\bar{c}, \bar{b})$ comprising two (not necessarily disjoint) sequences:

- $\bar{c} = \{c_\eta \in G \mid \eta \in \{0, 1\}^{<k}\}$, the set of *nodes* and
- $\bar{b} = \{b_\rho \in G \mid \rho \in \{0, 1\}^k\}$, the set of *branches*,

satisfying that for all $\eta \in \{0, 1\}^{<k}$ and $\rho \in \{0, 1\}^k$, if for some $\ell \in \{0, 1\}$ we have $\eta \smallfrown \langle \ell \rangle \triangleleft \rho$, then $b_\rho R c_\eta \equiv \ell$.

Sketch of the Proof of the Stable Regularity Lemma

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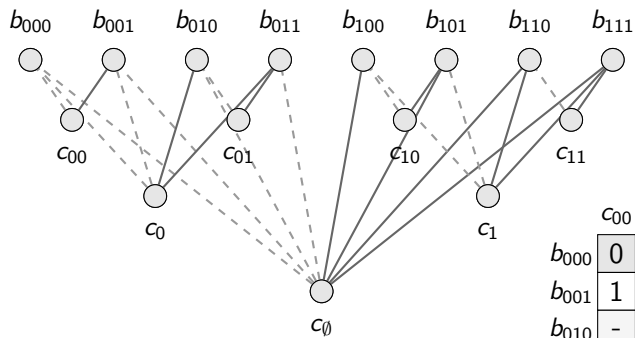
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Definition (Tree bound)

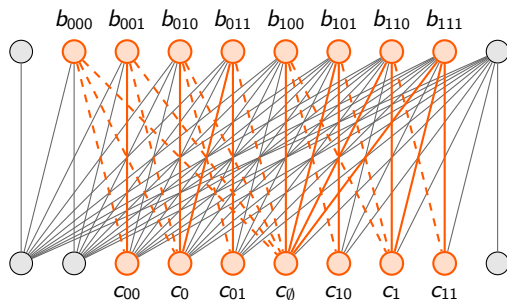
Suppose G is a finite graph. We denote the *tree bound* $k_{**} = k_{**}(G)$ as the minimal positive integer such that there is no k_{**} -tree $H = (\bar{c}, \bar{b})$ in G .

Sketch of the Proof of the Stable Regularity Lemma



	c_{00}	c_0	c_{01}	$c_{\emptyset}$	c_{10}	c_1	c_{11}
b_{000}	0	0	-	0	-	-	-
b_{001}	1	0	-	0	-	-	-
b_{010}	-	1	0	0	-	-	-
b_{011}	-	1	1	0	-	-	-
b_{100}	-	-	-	1	0	0	-
b_{101}	-	-	-	1	1	0	-
b_{110}	-	-	-	1	-	1	0
b_{111}	-	-	-	1	-	1	1

Sketch of the Proof of the Stable Regularity Lemma



	c_{00}	c_{01}	c_{10}	c_{11}						
	1	0	0	0	0	0	0	0	0	0
b_{000}	1	1	0	0	0	0	0	0	0	0
b_{001}	1	1	1	0	0	0	0	0	0	0
b_{010}	1	1	1	1	0	0	0	0	0	0
b_{011}	1	1	1	1	1	0	0	0	0	0
b_{100}	1	1	1	1	1	1	0	0	0	0
b_{101}	1	1	1	1	1	1	1	0	0	0
b_{110}	1	1	1	1	1	1	1	1	0	0
b_{111}	1	1	1	1	1	1	1	1	1	0
	1	1	1	1	1	1	1	1	1	1

Sketch of the Proof of the Stable Regularity Lemma

Theorem (Lemma 6.7.9 in [?])

*If a graph G has the $2^{k_{**}}$ -order property, then the tree bound of G is at least $k_{**} + 1$.*

Sketch of the Proof of the Stable Regularity Lemma

Theorem (Lemma 6.7.9 in [?])

*If a graph G has the $2^{k_{**}}$ -order property, then the tree bound of G is at least $k_{**} + 1$.*

*On the other hand, if a graph G has tree bound at least $k_{**} = 2^{k_*+1} - 1$, then it has the k_* -order property.*

Sketch of the Proof of the Stable Regularity Lemma

Lemma (Existence of excellent subsets, [?])

Let G be a finite graph with the non- k_ -order property. Let $\zeta \leq \frac{1}{2^{k_{**}}}$, $\epsilon \in (0, \frac{1}{2})$. Then, for every $A \subseteq G$ with $|A| \geq \frac{1}{\epsilon^{k_{**}}}$ there exists an (ϵ, ζ) -excellent subset $A' \subseteq A$ such that $|A'| \geq \epsilon^{k_{**}-1}|A|$.*

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TO DO

Sketch of the Proof of the Stable Regularity Lemma

$A_\emptyset$



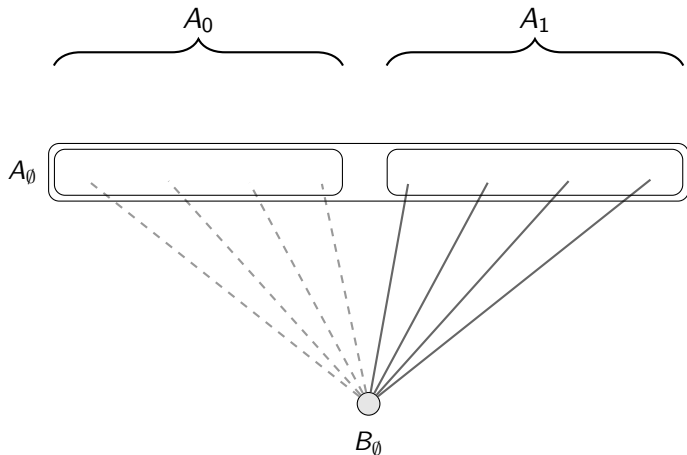
Sketch of the Proof of the Stable Regularity Lemma

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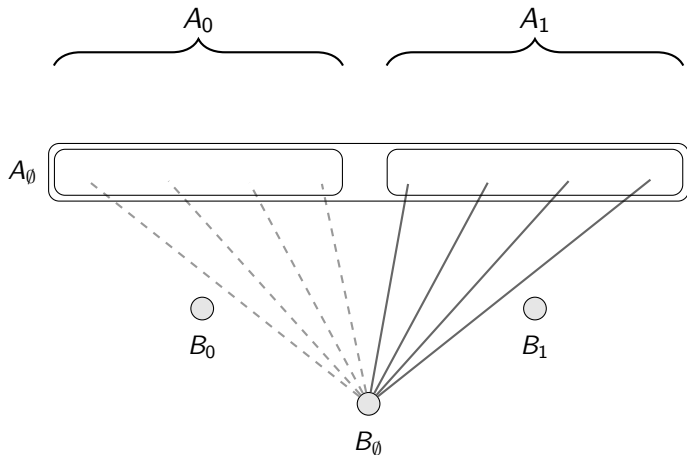


$B_\emptyset$

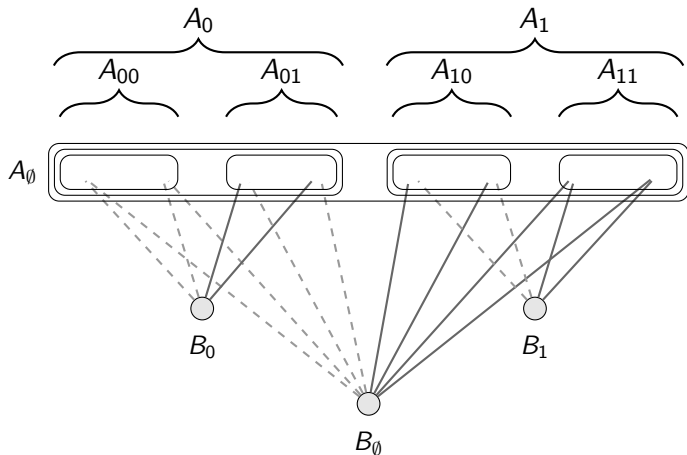
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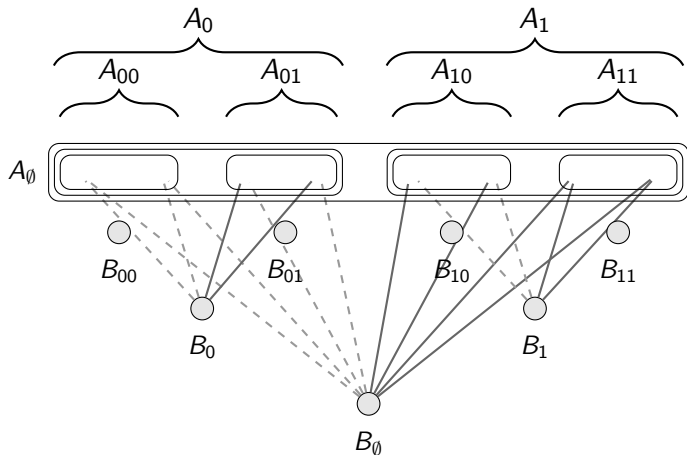
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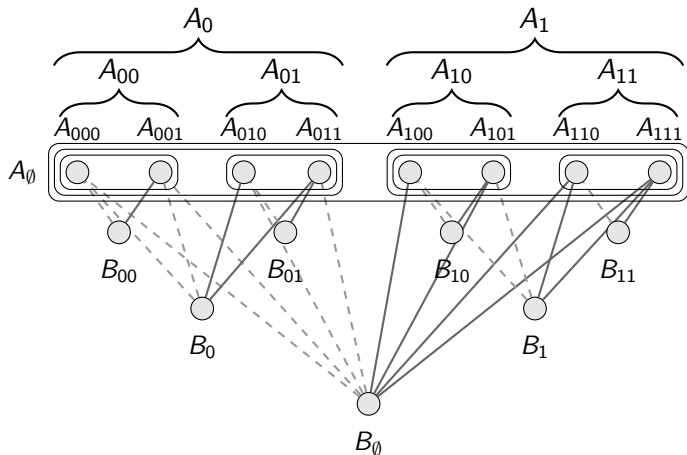
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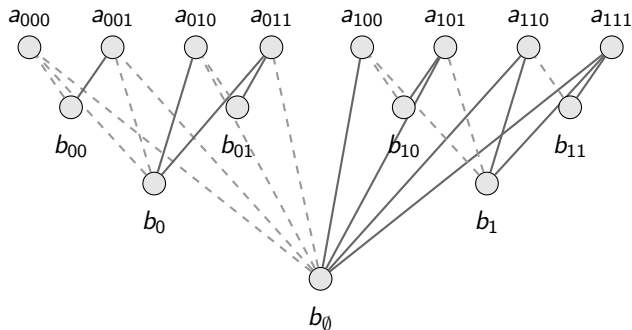
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Sketch of the Proof of the Stable Regularity Lemma

Lemma

Let G be a finite graph with the non- k_* -order property. Let $\epsilon \in (0, \frac{1}{2})$ and $\epsilon' \leq \frac{1}{2^{k_{**}}}$. Let $A \subseteq G$ such that $|A| = n \geq m_0$. Let $\langle m_\ell \mid \ell \in \{0, \dots, k_{**}\} \rangle$ be a decreasing sequence of natural numbers such that $\epsilon m_\ell \geq m_{\ell+1}$ for all $\ell \in \{0, \dots, k_{**} - 1\}$ and $m_{k_{**}} \geq 1$. Denote $m_* := m_0$ and $m_{**} := m_{k_{**}}$. Then, there is a partition $\bar{A} = \langle A_j \mid j \in \{1, \dots, j(*)\} \rangle$ with remainder $B = A \setminus \bigcup_{j < j(*)} A_j$ such that:

- ① For all $j \in \{1, \dots, j(*)\}$, $|A_j| \in \langle m_\ell \mid \ell \in \{0, \dots, k_{**} - 1\} \rangle$.
- ② For all $i \neq j \in \{1, \dots, j(*)\}$, $A_i \cap A_j = \emptyset$.
- ③ For all $j \in \{1, \dots, j(*)\}$, A_j is (ϵ, ϵ') -excellent.
- ④ $|B| < m_*$.

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Lemma

Let G be a finite graph with the non- k_ -order property. Then, for every $\epsilon \in (0, \frac{1}{2})$, $\zeta \in (0, \frac{1}{2} - \epsilon)$, $\xi \in (0, 1)$ and $m \geq \frac{1}{\zeta^2}(k_* \log \frac{1}{\zeta^2} k_* - \log \xi)$, if $A \subseteq G$ is an ϵ -good subset of size $n \geq m$, then a subset $A' \subseteq A$ of size m , sampled uniformly at random, is $(\epsilon + \zeta)$ -good with probability $1 - \xi$.*

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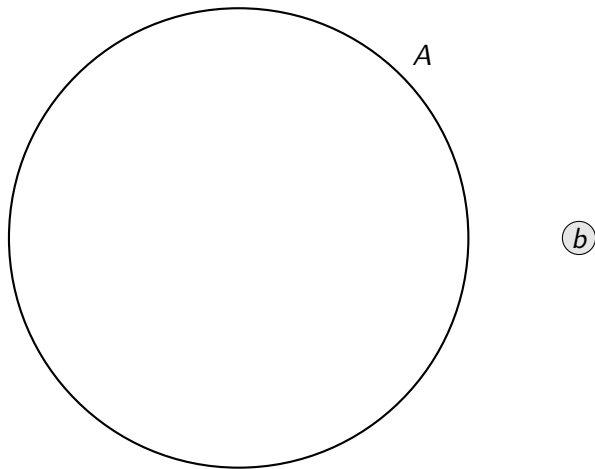
Lemma

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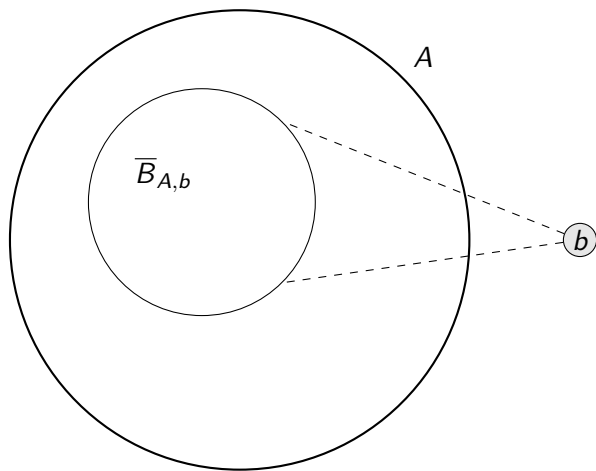
- $A \subseteq G$ is (ϵ, ϵ') -excellent and*
- $A' \subseteq A$ is $(\epsilon + \zeta')$ -good,*

then, A' is $(\epsilon + \zeta, \epsilon')$ -excellent.

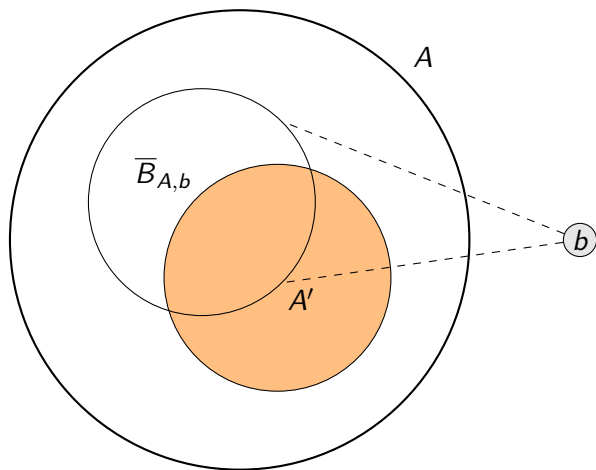
Sketch of the Proof of the Stable Regularity Lemma



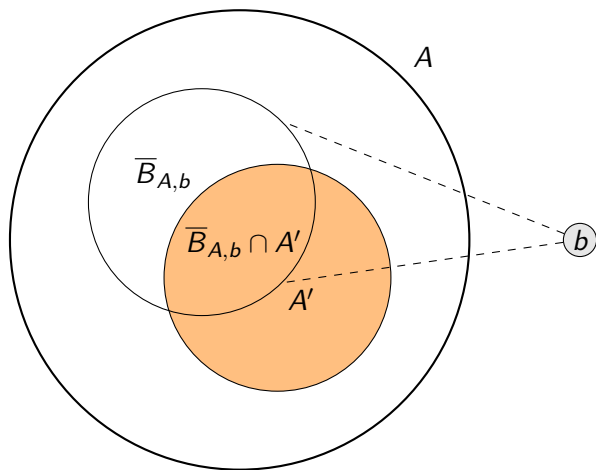
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Sketch of the Proof of the Stable Regularity Lemma

The probability of failure is small for each b :

$$\mathbb{P}(X_{A,b,A'}) \leq \mathbb{P} \left(|\overline{B}_{A,b} \cap A'| \geq \left(\frac{|\overline{B}_{A,b}|}{|A|} + \zeta \right) |A'| \right) \leq e^{-2\zeta^2 |A'|}$$

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Sketch of the Proof of the Stable Regularity Lemma

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Too many b 's?

Lemma

Let G be a graph with the (disjoint) non- k -order property. Then, for any finite non-trivial $A' \subseteq G$,

$$|\{\overline{B}_{A',b} \mid b \in G\}| \leq |A'|^k.$$

Sketch of the Proof of the Stable Regularity Lemma

Given the random variable

$$X = \sum_{\text{bad } \overline{B}_{A,b}} \mathbb{1}_{X_{A,b,A'}},$$

Sketch of the Proof of the Stable Regularity Lemma

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$$\mathbb{E}[X] = \sum_{\text{bad } \bar{B}_{A,b}} \mathbb{E}[\mathbb{1}_{X_{A,b,A'}}] = \sum_{\text{bad } \bar{B}_{A,b}} \mathbb{P}(X_{A,b,A'}) \leq \sum_{\text{bad } \bar{B}_{A,b}} e^{-2\zeta^2 m},$$

Sketch of the Proof of the Stable Regularity Lemma

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and by the First Moment Method

$$\mathbb{P}(X \geq 1) \leq \mathbb{E}[X] \leq m^{k_*} \cdot e^{-2\zeta^2 m} \leq \xi.$$

Sketch of the Proof of the Stable Regularity Lemma

Lemma

Let G be a finite graph with the non- k_ -order property. Then, for every $\zeta \in \{0, \frac{1}{2}\}$ and $\zeta' < \zeta$, there is $\epsilon_1 = \epsilon_1(\zeta, \zeta')$ such that for every $\epsilon < \epsilon' \leq \epsilon_1$, if*

- $A \subseteq G$ is (ϵ, ϵ') -excellent and*
- $A' \subseteq A$ is $(\epsilon + \zeta')$ -good,*

then, A' is $(\epsilon + \zeta, \epsilon')$ -excellent.

Proof idea: given a good set B , the number of exceptional edges between A and B , and thus between A' and B , is small, since A is (ϵ, ϵ') -excellent. Also, since B is good, each vertex in A' either has few or many exceptional edges with B . These two facts together imply that the number of vertices in A' that do not behave as the majority is small.

Sketch of the Proof of the Stable Regularity Lemma

Now, we know that random subsets of excellent sets are (slightly less) excellent with high probability. We can use this fact with a simple Union Bound argument to prove the following result.

Lemma

Let G be a finite graph with the non- k_ -order property. Then, for all $\zeta \in (0, \frac{1}{2})$, $\zeta' < \zeta$, $r \geq 1$ and for all $\epsilon < \epsilon'$ small enough ($\leq \epsilon_1(\zeta, \zeta')$ from the previous point) there exists $N = N(k_*, \zeta', r)$ such that: if $|A| = n > N$, r divides n and A is (ϵ, ϵ') -excellent, then there exists a partition into r disjoint pieces of equal size, each of which is $(\epsilon + \zeta, \epsilon')$ -excellent.*

Sketch of the Proof of the Stable Regularity Lemma

Applying this result to the previously obtained uneven partition, and carefully choosing some parameters, we obtain an even partition into excellent parts.

Lemma

Let G be a finite graph with the non- k_* -order property. Let ϵ' and ϵ be two real numbers such that $\epsilon' \leq \min\{\frac{1}{5}, \frac{1}{2^{k_{**}}}\}$ and $\epsilon \leq \frac{1}{4k}\epsilon'$ for some $k > 1$. Also, let m_* , m_{**} and q be natural numbers such that $q \geq \lceil \frac{1}{\epsilon} \rceil$, $m_{**} > \frac{N(k, k_*, \epsilon', \frac{m_*}{m_{**}})}{q}$ and $m_* := q^{k_{**}} m_{**}$. Then, for any $A \subseteq G$ with $|A| = n \geq m_*$ there exists a partition $\bar{A} = \langle A_i \mid i \in \{1, \dots, i(*)\} \rangle$ with remainder $B = A \setminus \bigcup \bar{A}$ such that:

- ① $i(*) \leq \frac{n}{m_{**}}$.
- ② For all $i \in \{1, \dots, i(*)\}$, $|A_i| = m_{**}$.
- ③ For all $i \in \{1, \dots, i(*)\}$, A_i is $(\frac{\epsilon'}{k}, \epsilon')$ -excellent.
- ④ $|B| < m_*$.

Sketch of the Proof of the Stable Regularity Lemma

This can be addressed by uniformly distributing the vertices of the remainder among the parts. This, again, slightly reduces the excellence of each part.

Lemma

Under the same condition of the previous lemma, we can get a partition $\bar{A} = \langle A_i \mid i \in \{1, \dots, i()\} \rangle$ with no remainder, such that:*

- ① *For all $i, j \in \{1, \dots, i(*)\}$, $||A_i| - |A_j|| \leq 1$.*
- ② *For all $i, j \in \{1, \dots, i(*)\}$, $A_i \cap A_j = \emptyset$.*
- ③ *For all $i \in \{1, \dots, i(*)\}$, A_i is (ϵ'', ϵ') -excellent, for some ϵ'' satisfying*

$$0 < \epsilon'' \leq \frac{\frac{\epsilon'}{k} m_{**} + \left\lceil \frac{m_*}{i(*)} \right\rceil}{m_{**} + \left\lceil \frac{m_*}{i(*)} \right\rceil}.$$

- ④ $A = \bigcup \bar{A}$.

Sketch of the Proof of the Stable Regularity Lemma

By carefully choosing hyperparameters, we can rephrase the result to only depend on the stability constant k_* , the desired excellence ϵ and the minimum partition size m .

Theorem

Let k_* be given. Then, for all $\epsilon \leq \min\{\frac{1}{5}, \frac{1}{2^{k_{**}}}\}$ and $m > 1$, there is $M = M(\epsilon, m, k_*)$ and $N = N(\epsilon, m, k_*)$ such that, for every finite graph G with the non- k_* -order property, and every $A \subseteq G$ with $|A| \geq N$, there exists a partition $\bar{A} = \langle A_i \mid i \in \{1, \dots, i(*)\} \rangle$ of A , such that:

- 1 The number of parts is bounded by $m \leq i(*) \leq M := \max\{\lceil \frac{12}{\epsilon} \rceil^{k_{**}+1}, 4m\}$.
- 2 For all $i, j \in \{1, \dots, i(*)\}$, $||A_i| - |A_j|| \leq 1$.
- 3 For all $i \in \{1, \dots, i(*)\}$, A_i is ϵ -excellent.

Sketch of the Proof of the Stable Regularity Lemma

Question: Why do we care about excellence?

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Lemma

For any $\epsilon_0, \epsilon \in (0, \frac{1}{2})$ such that $\epsilon_0 \leq \frac{\epsilon^2}{2}$, and for any two ϵ_0 -excellent sets A, B , we have that:

- 1 (A, B) is ϵ -regular.
- 2 If $A' \in [A]^{\geq \epsilon|A|}$ and $B' \in [B]^{\geq \epsilon|B|}$, then $d(A', B') < \epsilon$ or $d(A', B') \geq 1 - \epsilon$.

Sketch of the Proof of the Stable Regularity Lemma

Theorem (Stable Regularity Lemma, [?])

For every $k_* \in \mathbb{N}$ and $\epsilon \in (0, \frac{1}{2})$ and $m > 1$, there exist $N = N(\epsilon, m, k_*)$ and $M = M(\epsilon, m, k_*)$ such that, for every finite graph G with the non- k_* -order property, and every $A \subseteq G$ with $|A| \geq N$, the following holds. There exists $m < \ell < M$ and a partition $\bar{A} = \langle A_i \mid i \in \{1, \dots, \ell\} \rangle$ of A such that each A_i is $\frac{\epsilon^2}{2}$ -excellent and, for every (not necessarily distinct) $i, j \in \{1, \dots, \ell\}$,

- 1 $||A_i| - |A_j|| \leq 1$, i.e. the partition is equitable.
- 2 (A_i, A_j) is ϵ -regular, and moreover if $B_i \in [A_i]^{\geq \epsilon|A_i|}$ and $B_j \in [A_j]^{\geq \epsilon|A_j|}$, then either $d(B_i, B_j) < \epsilon$ or $d(B_i, B_j) \geq 1 - \epsilon$.
- 3 If $\epsilon \leq \min\{\frac{1}{5}, \frac{1}{2^{k_{**}}}\}$, then $M \leq \max\{\lceil \frac{12}{\epsilon} \rceil^{k_{**}+1}, 4m\}$.

Sketch of the Proof of the Stable Regularity Lemma

The fact that the pairs are either almost empty or almost full is not a core property of stable graphs, and is in fact proven in [?] to be satisfied by a larger class of graphs, the graphs with bounded *VC-dimension*. However, such graphs still require irregular pairs, something that stable graphs avoid.

Property Testing of H -freeness in Stable Graphs

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Conclusion and Future Work

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Thank you!

You can now rate the presentation. Was it *good*, *excellent* or *exceptional*?